



Handling Web Elements and Mouse Events

MAY, 2026

(expleo)

Contents

- Web Elements
- Web Elements ActionChains Class
- How to implement the ActionChains Class?

Web Elements



Web Elements

Introduction

- Anything that is present on the web page is a WebElement such as text box, button, etc.
- WebElement represents an HTML element.

Types of Web Element

- 1) Textbox
- 2) Button
- 3) Radio button
- 4) Checkbox
- 5) Dropdown
- 6) Link
- 7) Label



Web Elements ActionChains Class



ActionChains Class

- Actions are mainly triggered using two peripherals – Keyboard and Mouse.
- Actions like clicking a button, entering a keyword in the search bar are prime examples of how we use a mouse or keyboard.
- These interactions are automated by using the ActionChains class in Selenium.
- The ActionChains class contains a collection of actions that can be performed in a web application.
- Two broad categories are Keyboard and Mouse Actions.

ActionChains Class

1. Mouse Actions:

- **click():** Performs click on the element
- **double_click():** Performs double click on the element
- **click_and_hold():** Performs long click on the mouse without releasing it
- **drag_and_drop():** Drags the element from one point and drops to another
- **move_to_element():** Shifts the mouse pointer to the center of the element
- **context_click():** Performs right-click on the mouse

ActionChains Class

2. Keyboard Actions:

- **send_keys():** Sends a series of keys to the element
- **key_up():** Performs key release
- **key_down():** Performs keypress without release

Web Elements ActionChains Class

perform()

- perform method is used to perform all stored operations in action instance of ActionChains class.
- Syntax:

perform()

- Example:

```
<input type = "text" name = "passwd" id = "passwd-id" />
```

- To find an element one needs to use one of the locating strategies, For example,

```
element = driver.find_element_by_id("passwd-id")
```

```
element = driver.find_element_by_name("passwd")
```

Web Elements ActionChains Class

perform()

- Now one can use perform method as an Action chain as below:
- Create action chain object:

action = ActionChains(driver)

- Click the item:

action.click(on_element=element)

- Perform the operation:

action.perform()

How to implement the ActionChains Class?



How to implement the ActionChains Class?

Implementation Steps

- To implement the ActionChains class in Selenium automation script, follow the steps given below:

Step 1: First, we have to import the package

```
from selenium.webdriver import ActionChains
```

Step 2: To use the methods provided by the ActionChains class, we need to create an object of this class and pass the WebDriver as an argument.

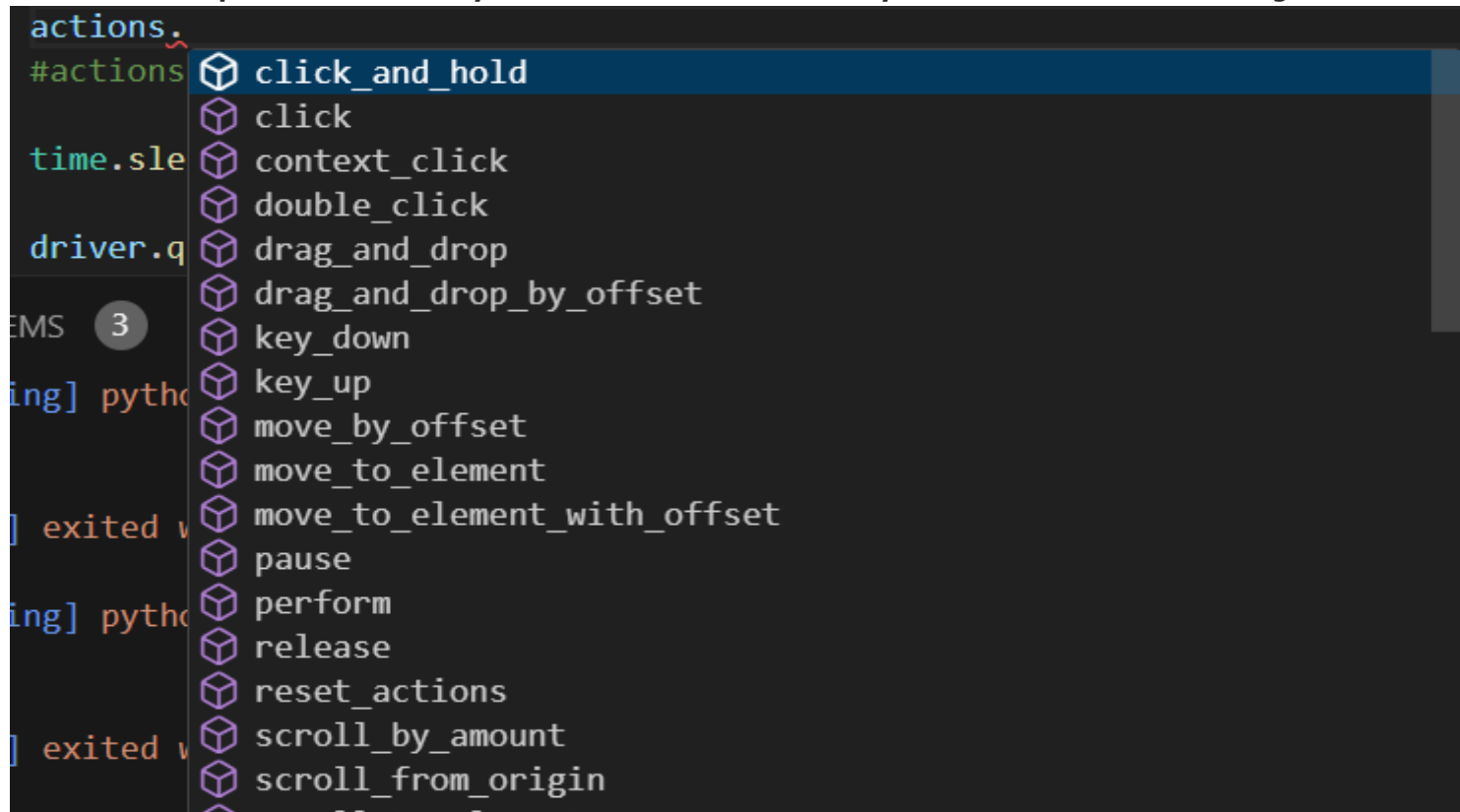
```
// instantiate the WebDriver  
WebDriver driver = new ChromeDriver();
```

```
// create an object of the Actions class  
action = ActionChains(driver);
```

How to implement the ActionChains Class?

Implementation Steps

Step 3: The object created can now be used to perform any actions. You can see various actions that are provided by this class once you create an object.



A screenshot of a Python IDE with a dark theme. On the left, a snippet of code is visible: `actions.`, `#actions`, `time.sleep`, `driver.quit`, and a terminal output showing `EMS 3`, `[ing] python`, `[ exited v`, `[ing] python`, and `[ exited v`. On the right, a dropdown menu is open, displaying a list of methods available for the `actions` object. Each method is preceded by a small cube icon. The methods listed are: `click_and_hold` (highlighted in blue), `click`, `context_click`, `double_click`, `drag_and_drop`, `drag_and_drop_by_offset`, `key_down`, `key_up`, `move_by_offset`, `move_to_element`, `move_to_element_with_offset`, `pause`, `perform`, `release`, `reset_actions`, `scroll_by_amount`, and `scroll_from_origin`.

How to Implement ActionChains class?

An Example Program using moveToElement(), click(), perform() methods

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver import ActionChains

driver = webdriver.Chrome()
driver.maximize_window()

driver.get("https://omayo.blogspot.com/")

actions = ActionChains(driver)

blogs = driver.find_element(By.ID, value="blogsmenu")
actions.move_to_element(blogs).perform()

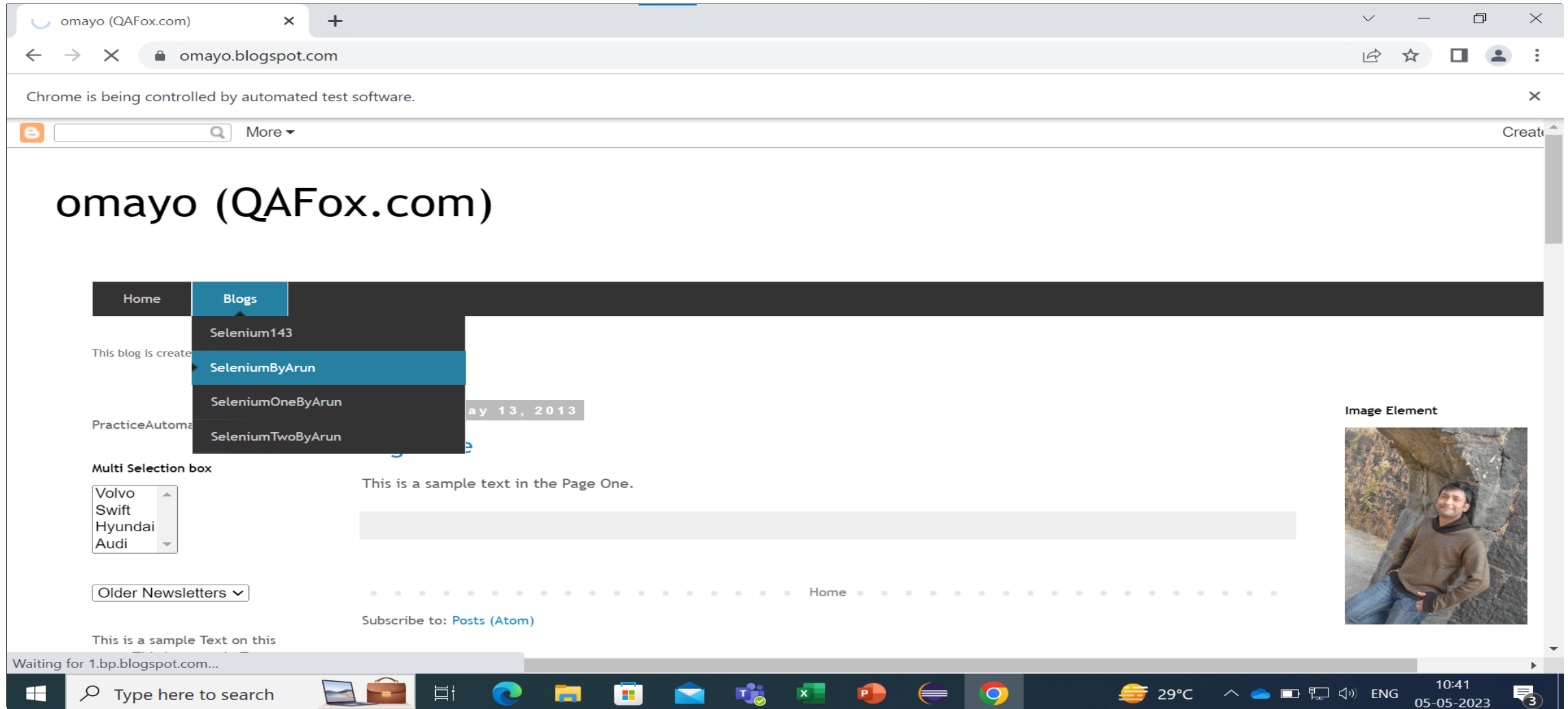
option2 = driver.find_element(By.XPATH, value="//span[text()='SeleniumByArun']")
actions.click(option2).perform()

time.sleep(5)

driver.quit()
```

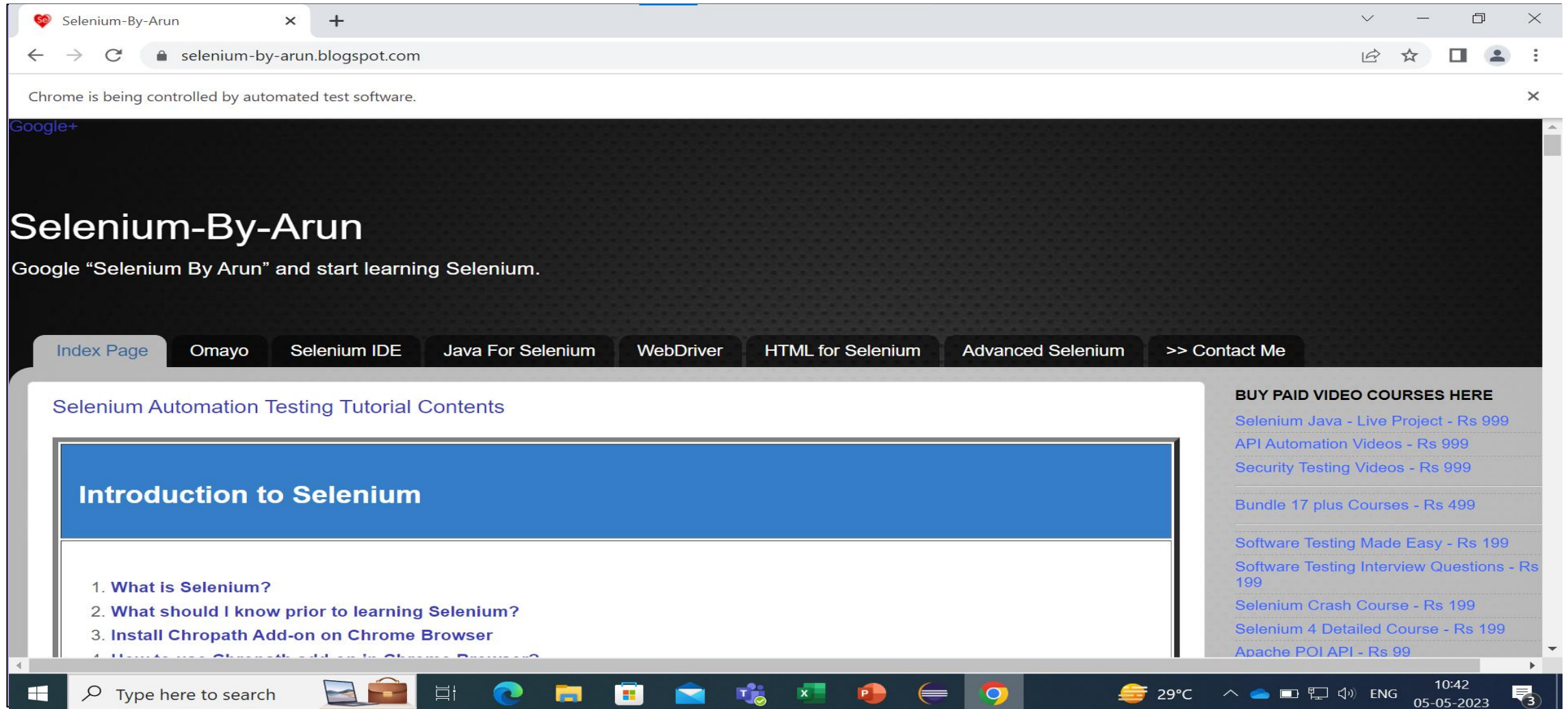
How to Implement ActionChains class?

Output



How to Implement ActionChains class?

Output



How to Implement ActionChains class?

An Example Program using context_click()

```
driver = webdriver.Chrome()
driver.maximize_window()

driver.get("https://omayo.blogspot.com/")

actions = ActionChains(driver)

#ActionChains Method - context_click(), perform()

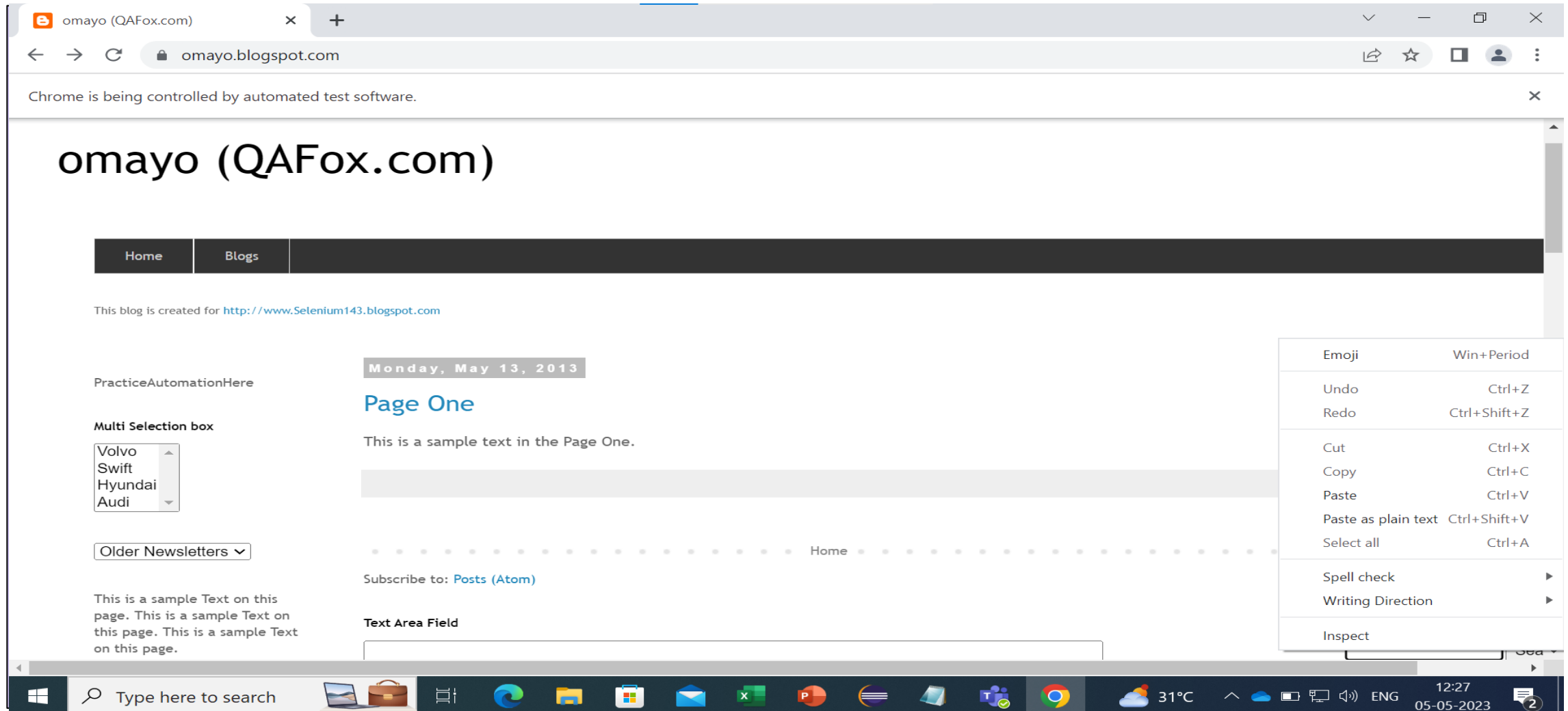
search_field = driver.find_element(By.XPATH, value="//input[@name='q']")
actions.context_click(search_field).perform()

time.sleep(5)

driver.quit()
```

How to Implement ActionChains class?

Output



How to Implement ActionChains class?

An Example Program using double_click() and perform()

```
driver = webdriver.Chrome()
driver.maximize_window()

driver.get("https://omayo.blogspot.com/")

actions = ActionChains(driver)

#ActionChains Method - double_click(), perform()

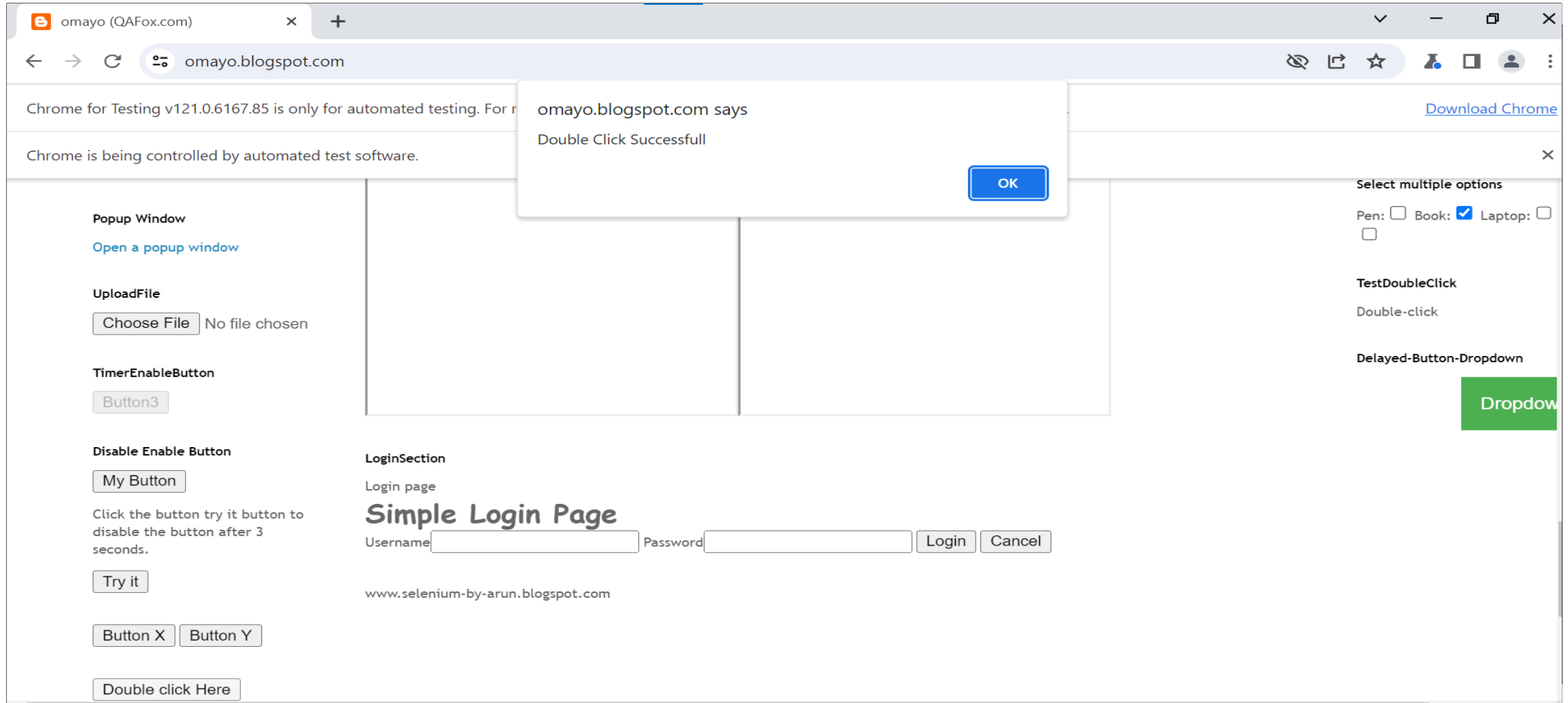
doubleclick = driver.find_element(By.XPATH, value="//button[text()=' Double click Here  ']")
actions.double_click(doubleclick).perform()

time.sleep(5)

driver.quit()
```

How to Implement ActionChains class?

Output



How to Implement ActionChains class?

An Example Program using `click_and_hold()`, `move_to_element()`, `release()` and `perform()`

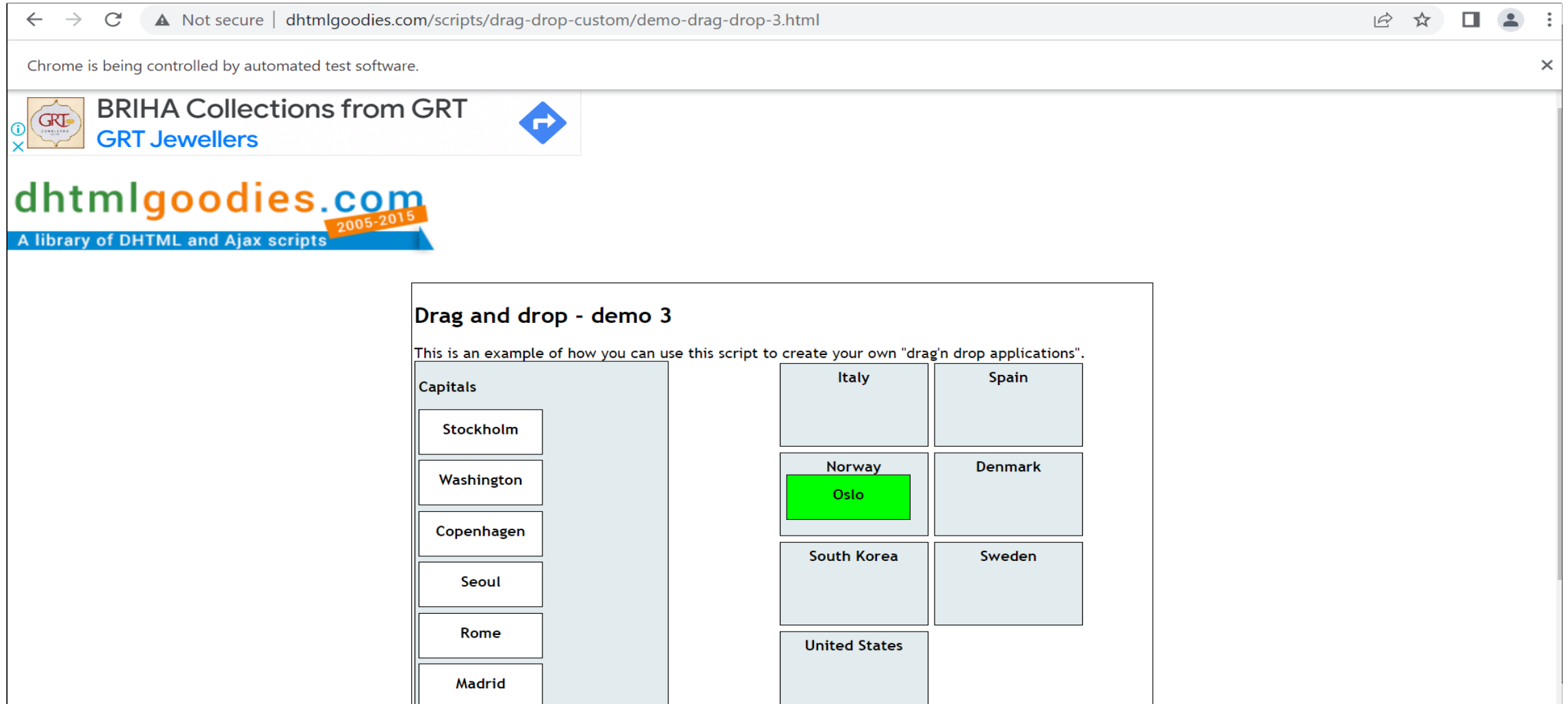
```
driver.get("http://dhtmlgoodies.com/scripts/drag-drop-custom/demo-drag-drop-3.html")
time.sleep(5)

box1 = driver.find_element(By.ID, value="box1")
box2 = driver.find_element(By.ID, value="box101")

actions.click_and_hold(box1).move_to_element(box2).release().perform()
time.sleep(5)
```

How to Implement ActionChains class?

Output



How to Implement ActionChains class?

An Example Program using drag_and_drop() and perform()

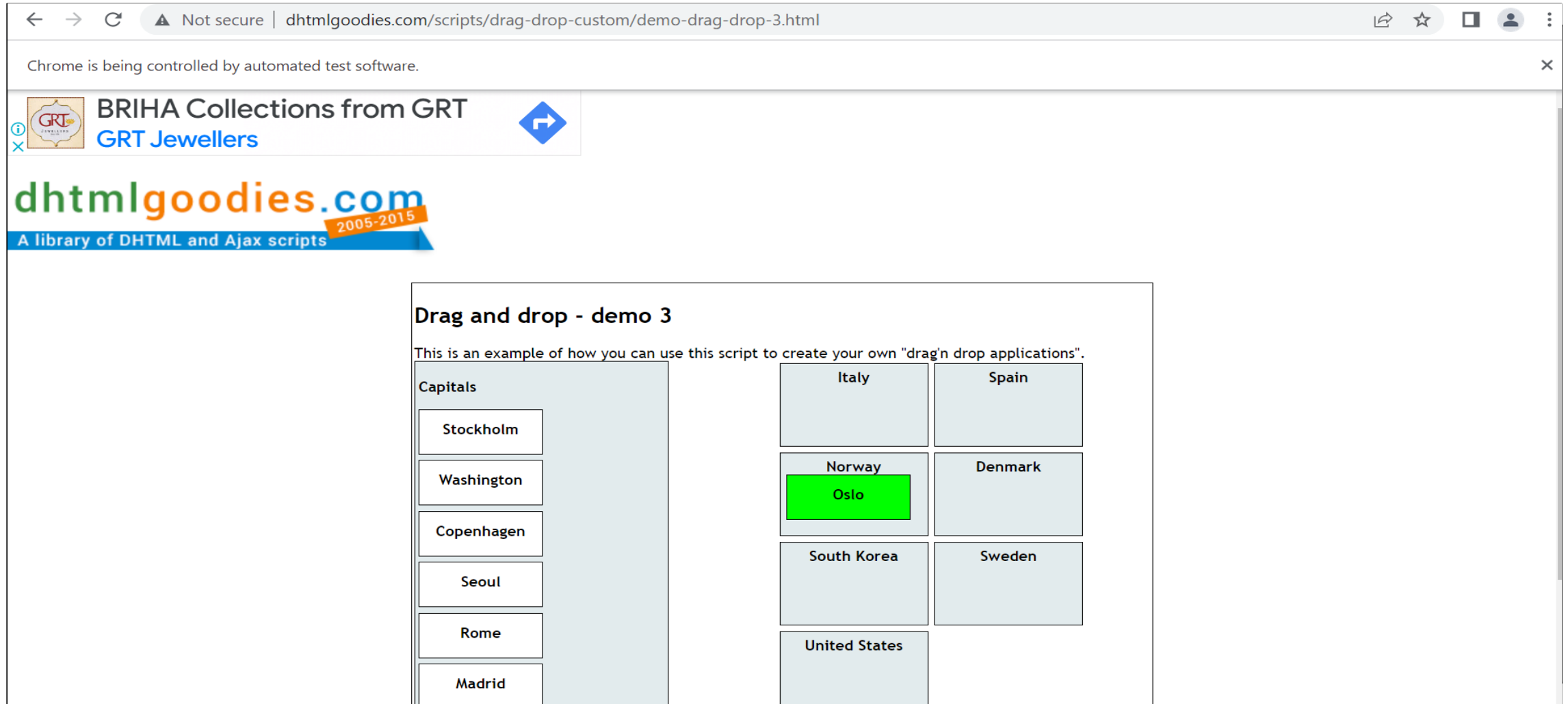
```
driver.get("http://dhtmlgoodies.com/scripts/drag-drop-custom/demo-drag-drop-3.html")
time.sleep(5)

box1 = driver.find_element(By.ID, value="box1")
box2 = driver.find_element(By.ID, value="box101")

actions.drag_and_drop(box1,box2).perform()
time.sleep(5)
```


How to Implement ActionChains class?

Output



How to Implement ActionChains class?

An Example Program using send_keys() and perform() methods

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver import ActionChains
from selenium.webdriver import Keys

driver = webdriver.Chrome()
driver.maximize_window()

action = ActionChains(driver)

driver.get("http://tutorialsninja.com/demo/index.php?route=account/login")

driver.find_element(By.ID, value="input-email").send_keys("aarun.83@gmail.com")

time.sleep(5)

driver.find_element(By.ID, value="input-password").send_keys("Arun123")

action.send_keys(Keys.ENTER).perform()
time.sleep(3)
driver.quit()
```

How to Implement ActionChains class?

Output

Output:

The screenshot displays a web browser window with the following elements:

- Browser Tab:** My Account
- Address Bar:** tutorialsninja.com/demo/index.php?route=account/account
- Browser Messages:**
 - Chrome for Testing v121.0.6167.85 is only for automated testing. For regular browsing, use a standard version of Chrome that updates automatically. [Download Chrome](#)
 - Chrome is being controlled by automated test software.
- Website Header:**
 - Currency: \$
 - Phone: 123456789
 - My Account
 - Wish List (0)
 - Shopping Cart
 - Checkout
- Search Bar:** Search
- Navigation Menu:** Desktops, Laptops & Notebooks, Components, Tablets, Software, Phones & PDAs, Cameras, MP3 Players
- Breadcrumb:** Home > Account
- Main Content:**
 - My Account**
 - [Edit your account information](#)
 - [Change your password](#)
 - [Modify your address book entries](#)
 - [Modify your wish list](#)
 - My Orders**
 - [View your order history](#)
 - [Downloads](#)
 - [Your Reward Points](#)
- Right Sidebar:**
 - My Account
 - Edit Account
 - Password
 - Address Book
 - Wish List
 - Order History
 - Downloads

How to Implement ActionChains class?

An Example Program using key_down(), key_up() and perform() methods

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver import ActionChains
from selenium.webdriver import Keys

driver = webdriver.Chrome()
driver.maximize_window()
driver.get("https://omayo.blogspot.com/")

links = driver.find_elements(By.XPATH, value="//div[@id='LinkList1']/a")
print(len(links))

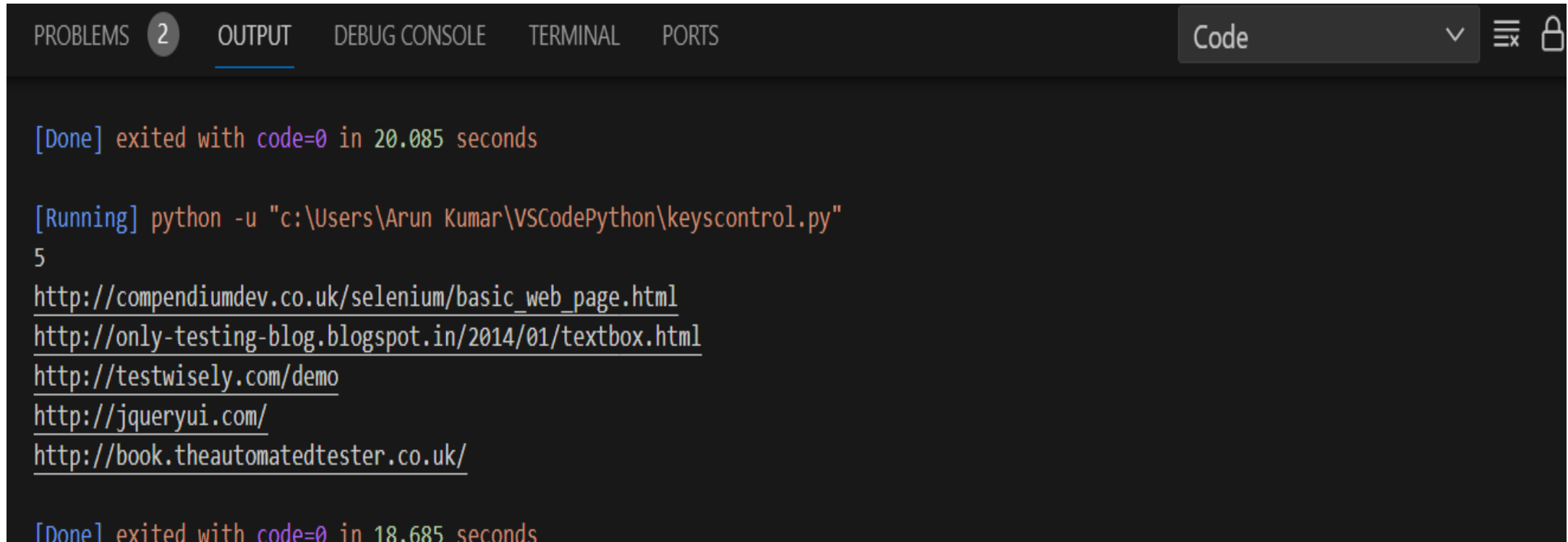
for link in links:
    print(link.get_attribute("href"))

for link in links:
    action = ActionChains(driver)
    action.key_down(Keys.CONTROL).click(link).key_up(Keys.CONTROL).perform()

time.sleep(5)
driver.quit()
```

How to Implement ActionChains class?

Output



The screenshot shows the VS Code Output window with the 'OUTPUT' tab selected. The output text is as follows:

```
[Done] exited with code=0 in 20.085 seconds

[Running] python -u "c:\Users\Arun Kumar\VSCodePython\keyscontrol.py"
5
http://compendiumdev.co.uk/selenium/basic_web_page.html
http://only-testing-blog.blogspot.in/2014/01/textbox.html
http://testwisely.com/demo
http://jqueryui.com/
http://book.theautomatedtester.co.uk/

[Done] exited with code=0 in 18.685 seconds
```

How to Implement ActionChains class?

Output

omayo (QAFox.com) x testpages.herokuapp x Only Testing: TextB x The page you were x jQuery UI x 500 Server Error x + - [] x

omayo.blogspot.com

Chrome for Testing v121.0.6167.85 is only for automated testing. For regular browsing, use a standard version of Chrome that updates automatically. [Download Chrome](#)

Chrome is being controlled by automated test software.

Enabled Button

Button2

Disabled Button

Button1

Disabled Text Box

Buttons with same name attribute values

Submit Login Register

Ordered List

- One
- Two
- Three
- Four
- Five
- Six

Table

Name	Age	Place
Kishore	22	Delhi
Manish	25	Pune
Praveen	29	Bangalore
Dheepthi	31	Mumbai

HTML Form

Username:

Password:

Login

Iframe1 & Iframe2

Readonly TextBox

ReadThisText

GetPrompt

GetPrompt

Confirmation Dialog

GetConfirmation

Hidden Button

Locate using name attribute

Other Sites to Practice Automat

- [compendiumdev](#)
- [onlytestingblog](#)
- [testwisely](#)
- [jqueryui](#)
- [theautomatedtester](#)

book.theautomatedtester.co.uk